Transformations of Polynomial Functions

Name _____

Algebra 2 Lesson 3-7 (fast track) Date _____ Period _____

Learning goal: I can describe how the graph of a cubic or quartic function is stretched, reflected, and translated from its parent function.

Language goal: I can explain each change to a graph using the words parent function, leading coefficient, stretch, reflection, and translation.

On the quiz: (1) Even, odd or neither from a graph (full Savvas quiz only). (2) The parent function and the ordered transformations of a quartic with a negative leading coefficient (Example 3, Practice). (3) Every true statement about a stretched and translated cubic (Example 3, Try It 3). (4, bonus) A quartic's equation from its turning point (Example 3 backwards). (5) A cube's volume in new units (full Savvas quiz only).

Example 3: Graph Transformations of Cubic and Quartic Parent Functions

How do transformed graphs compare to the graph of the parent function? The **parent functions** are $y = x^3$ (cubic) and $y = x^4$ (quartic). In $g(x) = a(x-h)^n + k$, the **leading coefficient** a stretches the graph vertically when $|a| > 1$, compresses it when $0 < |a| < 1$, and also reflects it across the x-axis when $a < 0$; h translates it horizontally (right when $h > 0$) and k vertically (up when $k > 0$).

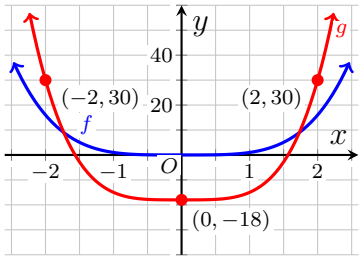
A. $g(x) = 3x^4 - 18$

1. The parent function is $f(x) = x^4$.

2. The leading coefficient, 3, **stretches** the graph vertically, making it narrower than the graph of the parent function.

3. Subtracting 18 **translates** the graph down 18 units.

on $f = x^4$	$\times 3$	-18: on g
$(0, 0)$	$(0, 0)$	$(0, -18)$
$(\pm 1, 1)$	$(\pm 1, 3)$	$(\pm 1, -15)$
$(\pm 2, 16)$	$(\pm 2, 48)$	$(\pm 2, 30)$



B. $h(x) = 5(x-2)^3 + 4$

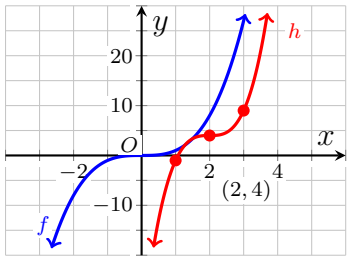
1. The parent function is $f(x) = x^3$.

2. Subtracting 2 (before calculating the cube) **translates** the graph right 2 units.

3. Multiplying by 5 **stretches** the translated graph vertically, making it narrower.

4. Adding 4 translates the stretched graph up 4 units.

on $f = x^3$	right 2	$\times 5$	$+4$: on h
$(-1, -1)$	$(1, -1)$	$(1, -5)$	$(1, -1)$
$(0, 0)$	$(2, 0)$	$(2, 0)$	$(2, 4)$
$(1, 1)$	$(3, 1)$	$(3, 5)$	$(3, 9)$



Check: $g(2) = 3(16) - 18 = 30$, $h(3) = 5 + 4 = 9$ ✓ **Order:** stretch or reflect *first*, then translate vertically.

Say it: The **parent function** of h is _____. The **leading coefficient** _____ _____ the graph vertically; the **translation** is _____ right and _____ up. No **reflection**: the leading coefficient is _____.

Try It 3: Compare to the parent function

How does the graph of the function $g(x) = 2x^3 - 5$ differ from the graph of its parent function?

Parent function: $y =$ _____

The leading coefficient, 2, _____

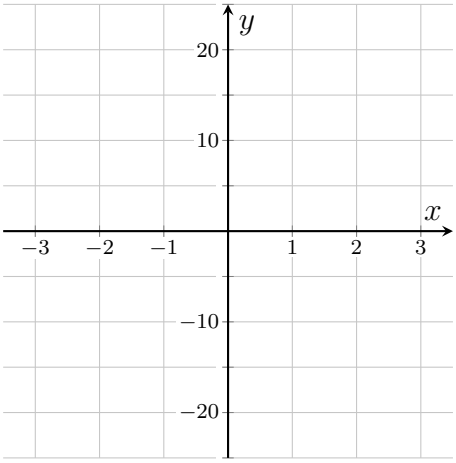
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Subtracting 5 _____

_____

Check one point: $(1, 1)$ on $y = x^3$ goes to _____ on g.

(Optional) Sketch the parent function and g on the grid.



Practice: When the leading coefficient is negative

A **negative** leading coefficient adds a **reflection** across the x-axis: the ends of the graph point the opposite way from the parent function's ends.

18. How do the graphs of transformed functions compare to the graph of the parent function?

$$g(x) = -x^4 - 8$$

Parent function: $y =$ _____

The leading coefficient of -1 _____

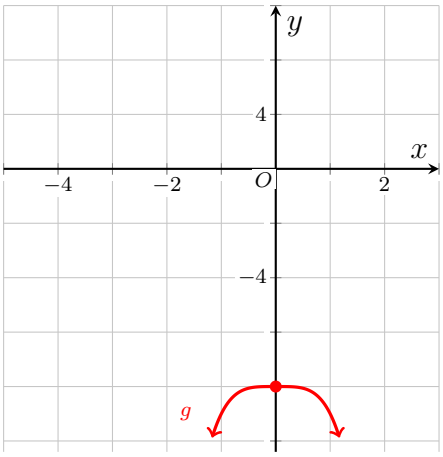
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Subtracting 8 _____

Which comes first, the reflection or the translation? Why?

_____

_____



8. Error Analysis Describe the error Terrence made in graphing the transformation of the cubic function $g(x) = x^3$ to $f(x) = -\frac{1}{2}x^3 + 10$.

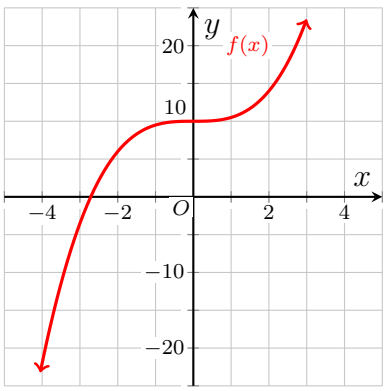
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Terrence's graph



Extension: Practice 12 (DOK 3), when you finish

An **even function** is symmetric about the y-axis: for all x, $f(x) = f(-x)$. An **odd function** is symmetric about the origin: for all x, $f(-x) = -f(x)$. The parent functions $y = x^2, x^4, x^6, \ldots$ are even; $y = x, x^3, x^5, \ldots$ are odd.

Construct Arguments Provide an example that demonstrates the following statement is not true.

If the degree of a function is an even number, then the function is an even function.

Your function: $f(x) =$ _____ Its degree: _____

Show that it is not an even function (use $f(-x)$, or explain what your transformation does to the symmetry about the y-axis):

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
KT8 ★	Describe function transformations	Quiz 2 and 3: name each change from the parent function
RSN ★	Transformations of functions	Quiz 2 and 4: the transformed graph and its equation
Q83	Even and odd functions	Optional: Quiz 1 (full Savvas quiz only) and the Extension
R7X	Function transformation rules (from 1-2)	Warm-up: a, h and k in $a\,f(x-h)+k$
XMK	Match polynomial functions and graphs using end behavior (from 3-1)	Warm-up: a negative leading coefficient flips the ends

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Before the quiz, finish these sentences.

In $g(x) = 4x^3 + 7$, the **parent function** is _____, the **leading coefficient** 4 _____, and adding 7 is a **translation** _____.

When the leading coefficient is negative, the graph has a **reflection** across the _____, so its ends _____.

To get $f(x) = -x^4 + 3$ from $y = x^4$, first _____, then _____, because _____.